Project Name

Blood Donor Finder

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WORKBOOK TEMPLATE

Project Name: Blood Donor Finder

1. Problem Statement

In emergencies like accidents, operations, or other medical cases, finding a suitable blood donor quickly can be difficult. Donor details are often shared through personal contacts, WhatsApp groups, or hospital notice boards, so it can take time to find someone who is available.

Our Blood Donor Finder is planned as a simple platform where donors can register, users can search for donors by blood group and city, and a blood request can be sent. The admin can verify details and update the request status.

2. Problem's reason

- **No Central Donor Database –** Donor information is available in different places, so it is difficult to search in one place.
- **Delay In Finding Donors –** Calling different people one by one takes time during an emergency.
- **Outdated Information –** Some donor numbers, locations, or availability details may no longer be correct.
- **Lack Of Verification –** It may be difficult for a seeker to know whether a donor is genuine and available.
- **Lack Of Tracking –** The user may not know whether a request is pending, accepted, or completed.
- **Poor Coordination –** Donors, users,and hospitals may not have a proper way to coordinate.
- **Lack Of Records –** There may be no proper record of requests and completed donations.

3. Internal Stakeholders

- **Blood Donor Finder Administrator –** manages users, donors, requests, status, and records more like a backbone.
- **Hospital / Blood Bank Staff –** helps coordinate donation and blood availability.
- **Organization/Institute Management –** looks after the overall system.
- **System Administrator/Developer –** maintains the website and database and prepare for future emergencies.

4. External Stakeholders

- **Blood Seekers –** These are the patients or families who search for donors when blood is needed.
- **Blood Donors –** These are the people who register and are willing to donate blood by their own choice.
- **Hospitals and Blood Banks –** It's receive donors and handle the donation process work like bridge between donor and seeker.
- **Community Members –** These are the peoples who help spread awareness and give feedback for increasing the uses.

5. Expected System Users

- **Administrator –** manages users, donors, requests, status, and records.
- **Donor –** Peoples willingly blood registers with blood group, city, and availability and responds to requests.
- **Seeker/User –** searches for donors and posts blood requests with required details.
- **Hospital/Blood Bank Officer –** handles requests related to their hospital or blood bank.

6. Objectives of the Project

- Develop an online blood donor finder system.
- Allow donors to register and update their availability.
- Allow seekers to search donors by blood group and location.
- Allow users to enter request details such as units needed, hospital, and urgency.
- Connect seekers with willing and verified donors.
- Manage and track blood requests.
- Show the current status of each request.
- Reduce the time needed to find a suitable donor.
- Maintain donor and donation records properly.

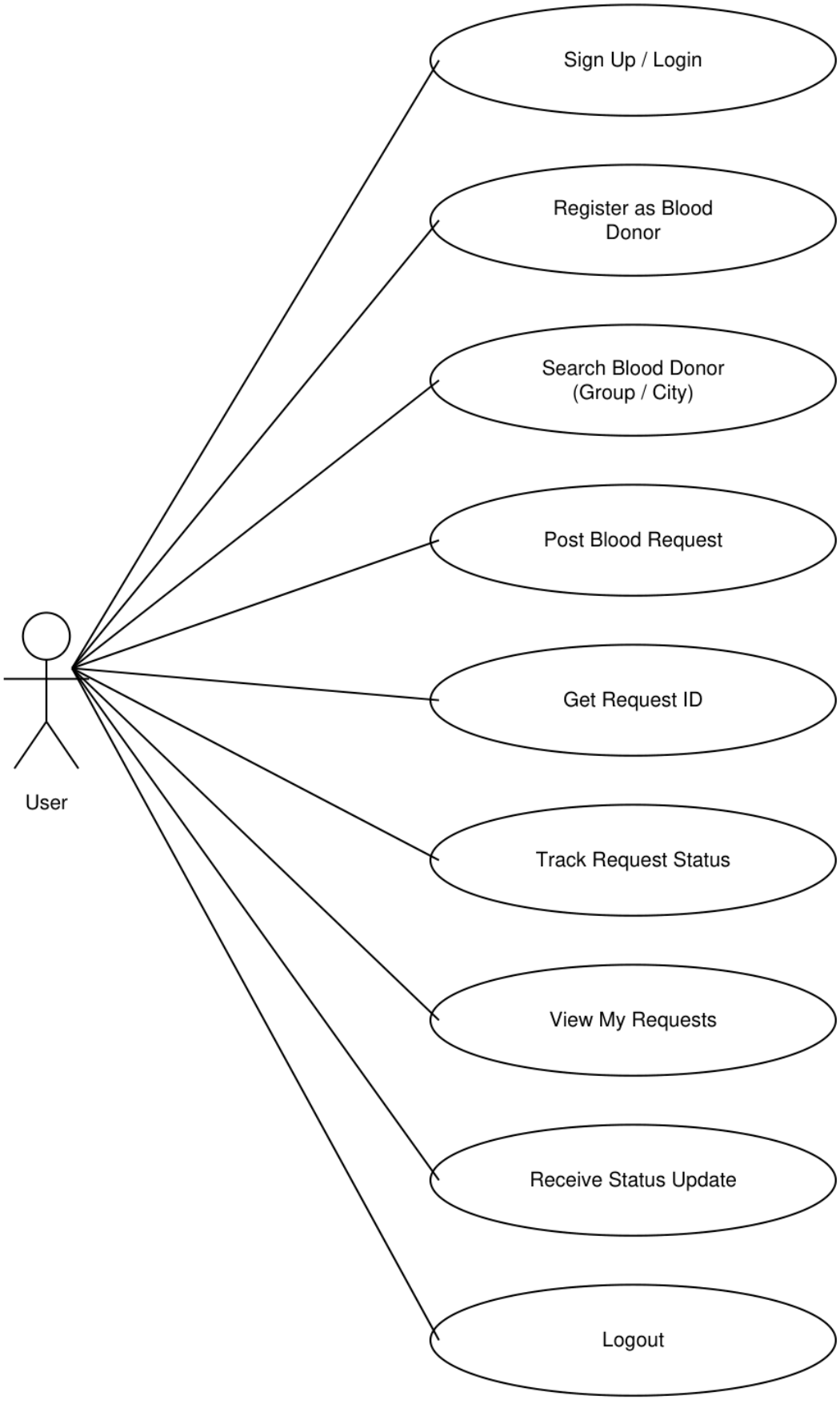
7. Expected Outputs

- List of available blood donors.
- Online blood request form.
- Request details with blood group, location, and hospital.
- Request confirmation with a Request ID.
- Request status such as Pending, In Progress, and Fulfilled.
- User request and donation history.
- Admin dashboard.
- Donor records and request reports.

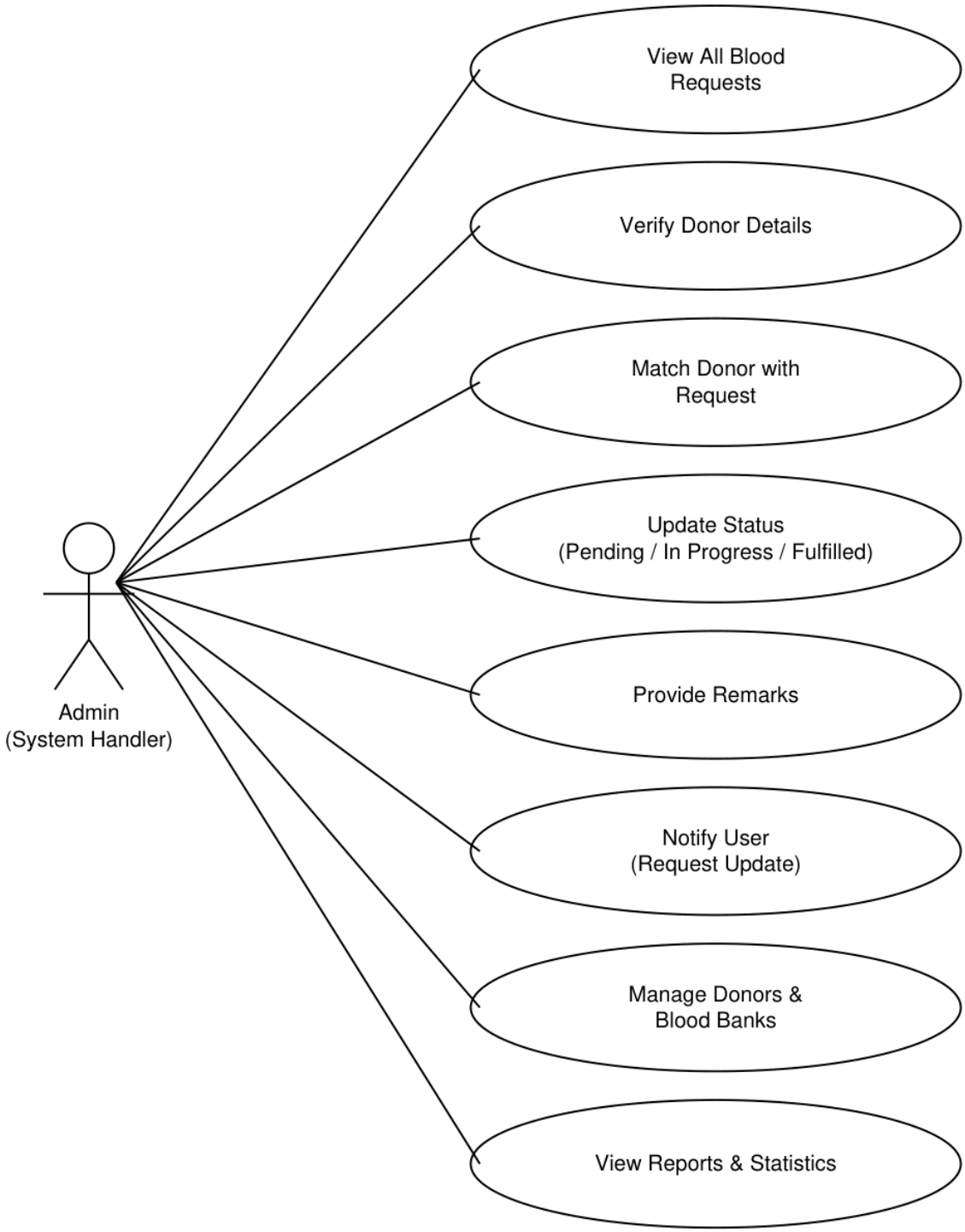
8. Hands-on Activity

- Find out how people currently look for blood donors during emergencies.
- Identify blood groups and common reasons for blood requests.
- Collect the details needed from donors and seekers.
- Prepare a simple data-gathering table.
- Design the website interface/wireframe.
- Prepare the blood request workflow.
- Identify stakeholders and expected system users.
- Design database tables for users, donors, requests, blood banks, and status.
- Prepare the system workflow for donor search, request, and fulfilment.

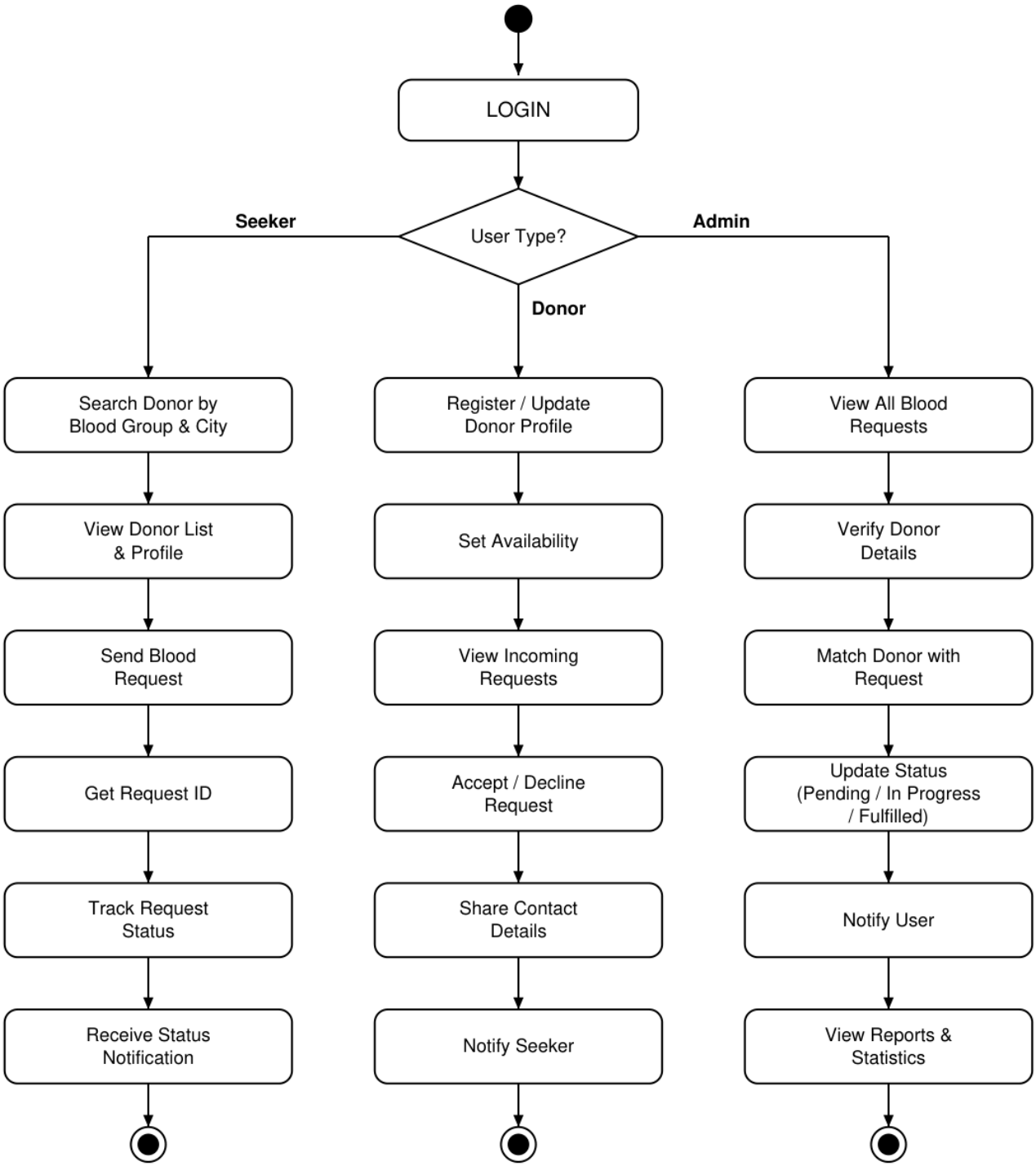
Use Case Diagram - User



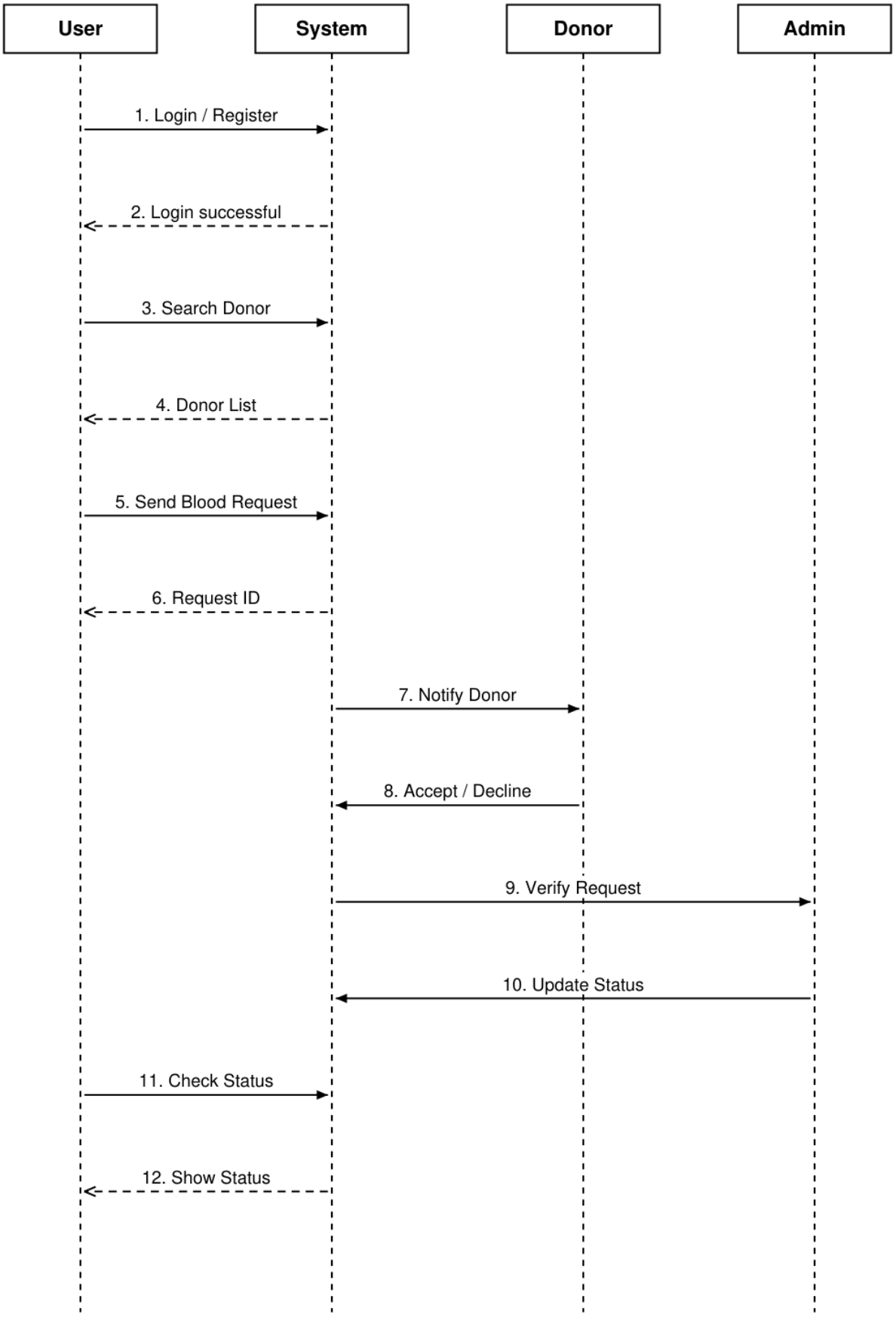
Use Case Diagram - Admin



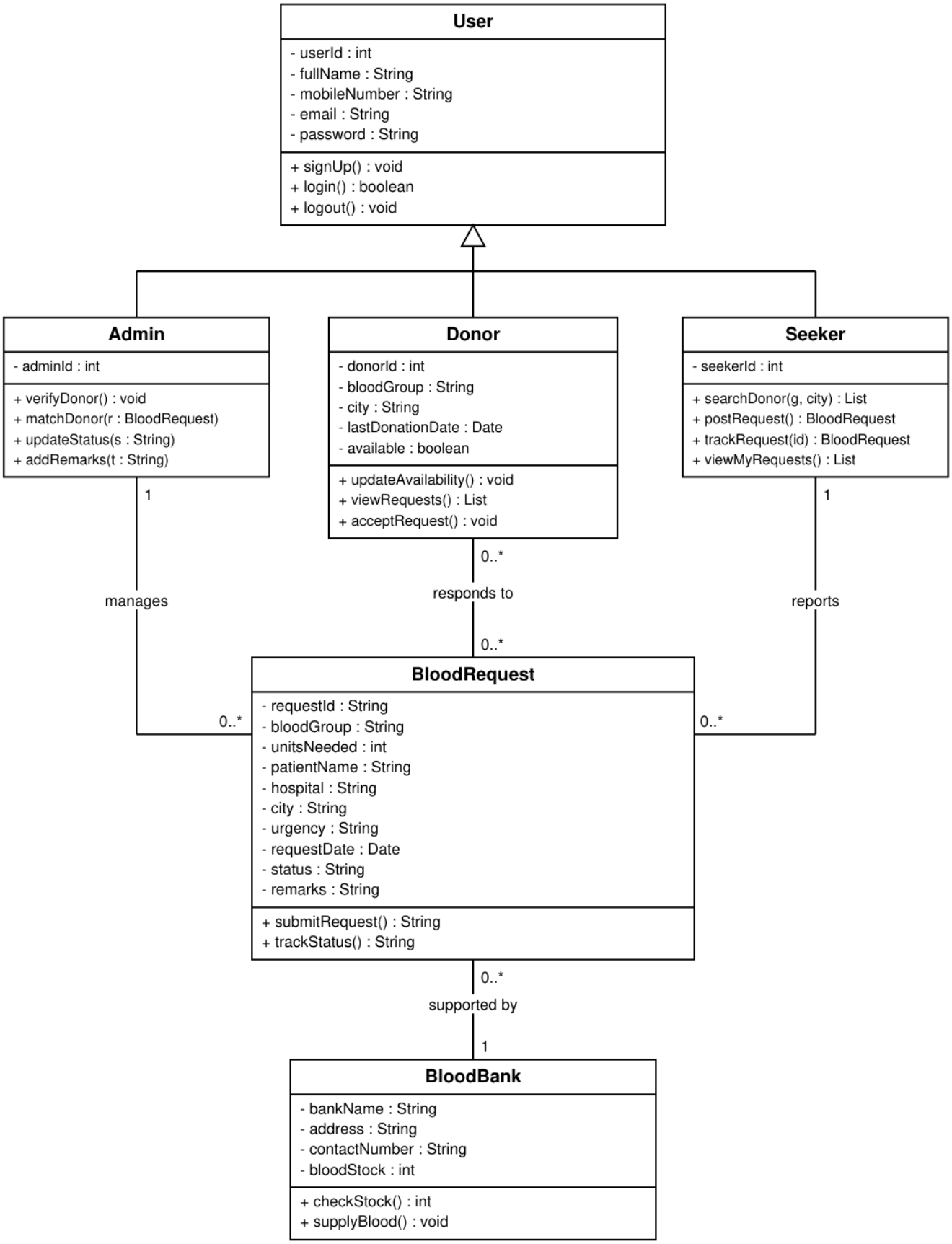
Blood Donor Finder - Activity Diagram



Blood Donor Finder Sequence Diagram



Blood Donor Finder - Class Diagram



Collaboration Diagram For Blood Donor Finder

